



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2007
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #2
(Year 12 Half Yearly)

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- All necessary working should be shown in every question.
- Hand in your answers in 3 separate bundles.
Section A (Question 1), Section B
(Question 2) and Section C (Question 3)

Total Marks – 60

- Attempt questions 1-3
- All questions are **NOT** of equal value.

Examiner: *A Ward*

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Section A – Start a new booklet.

Question 1 (20 marks).	Marks
a) The n^{th} term of an arithmetic series is $12 - 4n$. Find the first term, a and the common difference, d .	2
b) Differentiate with respect to x	
(i) $\cos 2x$	1
(ii) e^{x^2+1}	1
(iii) $\ln x^5$	1
(iv) $\frac{\sin x}{x^2}$	2
c) Convert $\frac{4\pi}{3}$ radians to degrees.	1
d) Express 210° in radian measure.	1
e) Find $\cos \theta$ and $\sin \theta$, if $\tan \theta = \frac{-7}{24}$ and θ is reflex.	2
f) Find:	
(i) $\int x^{2n} dx$ where $n \neq -\frac{1}{2}$	1
(ii) $\int \sec^2(2x+3) dx$	1
g) Find the gradient of the curve $y = (x-3)(x^2+2)$ at the point (1,-6)	2
h) Find the length of the arc of a circle of radius 4cm which subtends an angle of $\frac{\pi}{3}$ radians at the centre.	1
i) Find the equation of the tangent to the curve $y = x^2 - 3x + 2$ at the point where it cuts the y-axis.	1
j) The co-ordinates of the vertices A, B and C of the triangle ABC are (-3,7), (2,19) and (10,7) respectively. Prove that the triangle is isosceles.	3

End of Section A.

Section B – Start a new booklet.

Question 2 (20 marks).	Marks
a) The gradient function of a curve is $3x^2 - 5x + 1$ and the curve itself passes through (0,3). Find the equation of the curve.	2
b) Find:	
(i) $\int e^{3x+2} dx$	1
(ii) $\int \frac{\cos x}{1 + \sin x} dx$	1
c) Use Simpson's rule with 3 ordinates to find an approximate value for	
$\int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{x} dx.$	3
Express your answer as a fraction.	
d) Given the curve with equation $y = x^3 + 5x^2 + 3x - 9$, find the turning points and determine their nature.	3
e) Find the sum of geometric series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{4096}$	2
f) The curve $y = x^2 + ax + b$ has a turning point at (1,3). Find a and b .	2
g) If a and b are the first and last terms of an arithmetic series of $r + 2$ terms. Find the second and $(r+1)^{\text{th}}$ term, expressing your answers as single fractions.	3
h) Show that the point A(-1,5) is the reflection (image) of point B(3,-3) in the line $2y = x + 1$.	3

End of Section B.

Section C – Start a new booklet.**Question 3 (20 marks).****Marks**

- a)**
- (i) Sketch on the same axes the graphs of $y = -\cos 2x$ and $y = \frac{x}{2}$ in the interval, $-\pi \leq x \leq \pi$ 3
- (ii) Hence, find the number of solutions to the equation $-\cos 2x = \frac{x}{2}$, lying in the above interval. 2
- b)** A region R in the first quadrant is bounded by the y-axis, x-axis, the line $x = 3$ and the curve $y^2 = 4 - x$
- (i) Draw a sketch showing the region R. 2
- (ii) Calculate the area of region R. 3
- (iii) Calculate the volume formed when R is rotated about the y-axis through one revolution. 3
- c)** An insurance policy pays the policy holder a percentage of his salary if he is unable to work. For the first month the payment is 100% of his salary, for the second month 97% and for the third month 94.15%. The percentages are calculated according to the formula:
- $$P_{n+1} = aP_n + b$$
- Where P_n is the percentage paid in month n , with a and b as constants.
- (i) Find a and b . 2
- (ii) Show that $P_3 = 100(0.95)^2 + 2(1 + 0.95)$ and that $P_n = 100(0.95)^{n-1} + 2(1 + 0.95 + \dots + 0.95^{n-2})$, where $n \geq 2$ 2
- (iii) Hence, given that the sequences of the percentages tends to a limit P , find the value of P . 3

End of Section C.**End of Examination.**

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$

$$i) a) T_n = 12 - 4n$$

$$T_1 = 12 - 4(1)$$

$$T_1 = 8$$

$$T_2 = 12 - 4(2)$$

$$T_2 = 4$$

$$a = 8$$

$$d = T_2 - T_1$$

$$d = 4 - 8$$

$$d = -4$$

$$b) i) y = \cos 2x$$

$$y' = -2 \sin 2x$$

$$ii) y = e^{x^2+1}$$

$$y' = 2x e^{x^2+1}$$

$$iii) y = \ln x^5$$

$$y = 5 \ln x$$

$$y' = \frac{5}{x}$$

$$iv) y = \frac{\sin x}{x^2} \quad \begin{matrix} u = \sin x & v = x^2 \\ u' = \cos x & v' = 2x \end{matrix}$$

$$y' = \frac{vu' - uv'}{v^2}$$

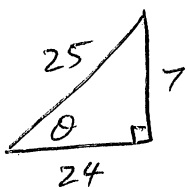
$$y' = \frac{x^2 \cos x - 2x \sin x}{x^4}$$

$$y' = \frac{x \cos x - 2 \sin x}{x^3}$$

$$c) \frac{4\pi}{3} \times \frac{180}{\pi} = 240^\circ$$

$$d) 210 \times \frac{\pi}{180} = \frac{7\pi}{6}$$

e)



$$\cos \theta = \frac{24}{25}$$

$$\sin \theta = -\frac{7}{25}$$



$$f) i) \int x^{2n} dx, \text{ where } n \neq -\frac{1}{2}$$

$$= \frac{x^{2n+1}}{2n+1} + C$$

$$ii) \int \sec^2(2x+3) dx$$

$$= \frac{1}{2} \tan(2x+3) + C$$

$$g) y = (x-3)(x^2+2)$$

$$y = x^3 + 2x - 3x^2 - 6$$

$$y' = 3x^2 + 2 - 6x$$

when $x = 1$

$$m_T = 3(1)^2 + 2 - 6(1)$$

$$m_T = -1$$

$$h) \ell = r\theta$$

$$\ell = 4\left(\frac{\pi}{3}\right)$$

$$\ell = \frac{4\pi}{3} \text{ cm}$$

$$i) y = x^2 - 3x + 2$$

$$y' = 2x - 3$$

when $x = 0$

$$m_T = -3 \quad \text{point } (0, 2)$$

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -3(x - 0)$$

$$y - 2 = -3x$$

$$y = -3x + 2 \text{ or } 3x + y - 2 = 0$$

$$j) A(-3, 7), B(2, 19), C(10, 7)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$AB = \sqrt{(2+3)^2 + (19-7)^2}$$

$$= 13$$

$$BC = \sqrt{(2-10)^2 + (19-7)^2}$$

$$= \sqrt{208}$$

$$AC = \sqrt{(10+3)^2 + (0)^2}$$

$$= 13$$

since $AB = AC$

ΔABC is isosceles.

2007 Mathematics Assessment 2: **Solutions— Section B**

2. (a) The gradient function of a curve is $3x^2 - 5x + 1$ and the curve itself passes through $(0, 3)$. Find the equation of the curve. [2]

Solution: $\frac{dy}{dx} = 3x^2 - 5x + 1,$
 $y = x^3 - \frac{5x^2}{2} + x + 3.$
 At $(0, 3)$ $3 = c,$
 $\therefore y = x^3 - \frac{5x^2}{2} + x + 3.$

- (b) Find:

i. $\int e^{3x+2} dx,$ [1]

Solution: $\int e^{3x+2} dx = \frac{e^{3x+2}}{3} + c.$

ii. $\int \frac{\cos x}{1 + \sin x} dx.$ [1]

Solution: $\int \frac{\cos x}{1 + \sin x} dx = \ln(1 + \sin x) + c.$

- (c) Use Simpson's rule with 3 ordinates to find an approximate solution for [3]

$$\int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{x} dx.$$

Express your answer as a fraction.

Solution: $\int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{x} dx \approx \frac{1}{3} \times \frac{1}{2} \left(2 + 4 \times 1 + \frac{2}{3} \right),$
 $\approx \frac{1}{6} \times \frac{20}{3},$
 $\approx \frac{10}{9}.$

- (d) Given the curve with equation $y = x^3 + 5x^2 + 3x - 9$, find the turning points and determine their nature. [3]

Solution: $\frac{dy}{dx} = 3x^2 + 10x + 3,$
 $= (3x + 1)(x + 3),$
 $= 0 \text{ when } x = -\frac{1}{3}, -3.$

$$\frac{d^2y}{dx^2} = 6x + 10,$$

$$= 8 \text{ when } x = -\frac{1}{3},$$

$$= -8 \text{ when } x = -3.$$

$\therefore$ Maximum turning point at $(-3, 0).$

Minimum turning point at $(-\frac{1}{3}, -10\frac{16}{27})$ or $(-\frac{1}{3}, -\frac{286}{27}).$

- (e) Find the sum of the geometric series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{4096}.$

2

Solution: $a = 1,$
 $r = \frac{1}{2},$
 $(\frac{1}{2})^{n-1} = (\frac{1}{2})^{12},$
 $\therefore n = 13.$

$$\text{So sum} = \frac{1 - (\frac{1}{2})^{13}}{1 - \frac{1}{2}},$$

$$= 2 \times \left(\frac{8192 - 1}{8192} \right),$$

$$= \frac{8191}{4096}.$$

- (f) The curve $y = x^3 + ax + b$ has a turning point at $(1, 3).$
Find a and $b.$

2

Solution: $y = x^3 + ax + b,$

i.e. $3 = 1 + a + b,$

$a + b = 2.$

$$\frac{dy}{dx} = 2x + a,$$

i.e. $0 = 2 + a,$

$a = -2.$

$-2 + b = 2,$

$b = 4.$

- (g) If a and b are the first and last terms of an arithmetic series of $r + 2$ terms, find the second and $(r + 1)^{\text{th}}$ terms, expressing your answers as single fractions. 3

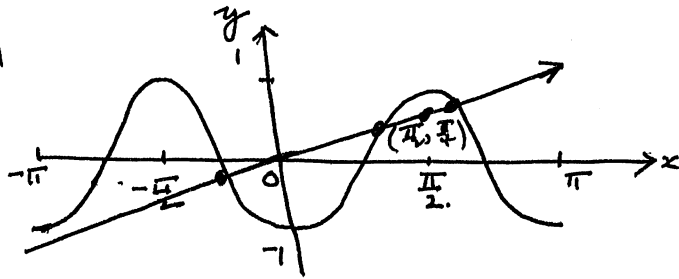
Solution: $t_1 = a, \quad t_{r+2} = b,$
i.e. $b = a + (r + 2 - 1)d,$
 $\therefore d = \frac{b - a}{r + 1}.$
 $t_2 = a + \frac{b - a}{r + 1},$
 $= \frac{ar + a + b - a}{r + 1},$
 $= \frac{ar + b}{r + 1}.$
 $t_{r+1} = b - \frac{b - a}{r + 1},$
 $= \frac{br + b - b + a}{r + 1},$
 $= \frac{a + br}{r + 1}.$

- (h) Show that the point $A(-1, 5)$ is the reflection (image) of point $B(3, -3)$ in the line $2y = x + 1$. 3

Solution: Slope of $2y = x + 1$ is $\frac{1}{2}$.
Slope of $AB = \frac{5 + 3}{-1 - 3},$
 $= -2.$
Distance of A from the line $= \frac{|-1 - 2(5) + 1|}{\sqrt{1 + 2^2}},$
 $= \frac{10}{\sqrt{5}}.$
Distance of B from the line $= \frac{|3 - 2(-3) + 1|}{\sqrt{1 + 4}},$
 $= \frac{10}{\sqrt{5}}.$
 $\therefore$ The line is the perpendicular bisector of $AB,$
i.e. A is the reflection of B in the line.

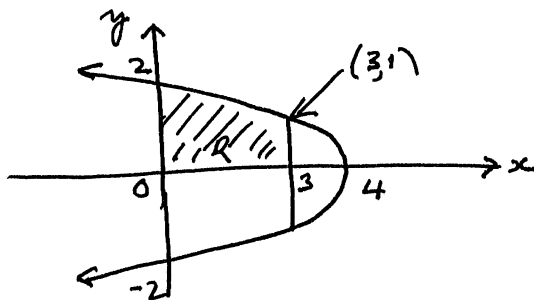
QUESTION 3.

(a) (i)



(ii) 3

(b) (i)



$$\begin{aligned}
 (ii) \quad & \int_0^3 \sqrt{4-x} \, dx \\
 &= \int_0^3 (4-x)^{\frac{1}{2}} \, dx \\
 &= -\frac{2}{3} \left[(4-x)^{\frac{3}{2}} \right]_0^3 \\
 &= -\frac{2}{3} \left(1^{\frac{3}{2}} - 4^{\frac{3}{2}} \right) \\
 &= -\frac{2}{3} (1 - 8) \\
 &= \frac{14}{3} \text{ m}^2.
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad V &= \pi \int_0^1 3^2 \, dy + \pi \int_1^2 x^2 \, dy \\
 &= \pi \int_0^1 9 \, dy + \pi \int_1^2 (4-y^2)^2 \, dy \\
 &= \pi [9y]_0^1 + \pi \int_1^2 (16 - 8y^2 + y^4) \, dy \\
 &= 9\pi + \pi \left[16y - \frac{8y^3}{3} + \frac{y^5}{5} \right]_1^2 \\
 &= 9\pi + \pi \left[32 - \frac{64}{3} + \frac{32}{5} - \left(16 - \frac{8}{3} + \frac{1}{5} \right) \right] \\
 &= 9\pi + \pi \left[16 - \frac{56}{3} + \frac{31}{5} \right] \\
 &= 9\pi + \pi \frac{240 - 280 + 93}{15} \\
 &= 9\pi + \frac{53\pi}{15} \\
 &= \frac{188}{15} \pi \text{ m}^3.
 \end{aligned}$$

$$\underline{c} \quad (i) \quad 97 = 100a + b \quad \text{--- (1)}$$

$$94.15 = 97a + b \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2}$$

$$2.85 = 3a$$

$$\boxed{a = 0.95}$$

Substitute in (1)

$$97 = 100 \times 0.95 + b$$

$$97 = 95 + b.$$

$$\therefore \boxed{b = 2.}$$

$$(ii) \quad P_2 = 100 \times 0.95 + 2.$$

$$P_3 = P_2 \times 0.95 + 2$$

$$= (100 \times 0.95 + 2) \times 0.95 + 2.$$

$$= 100 (0.95)^2 + 2 \times 0.95 + 2$$

$$= 100 (0.95)^2 + 2 (1 + 0.95)$$

$$P_4 = 100 (0.95)^3 + 2 (1 + 0.95 + 0.95^2)$$

etc.

$$P_n = 100 (0.95)^{n-1} + 2 (1 + 0.95 + 0.95^2 + \dots + 0.95^{n-2})$$

which is true
for $n \geq 2$.

$$(iii) \quad \text{Now } P_n = 100 (0.95)^{n-1} + \frac{2 (1 - 0.95^{n-1})}{1 - 0.95}.$$

$$P_n = 100 (0.95)^{n-1} + \frac{2 (1 - 0.95^{n-1})}{1/20}.$$

$$= 100 (0.95)^{n-1} + 40 (1 - 0.95^{n-1})$$

$$\text{now as } n \rightarrow \infty \quad 0.95^{n-1} \rightarrow 0 \quad \text{and } P_n \rightarrow \boxed{P = 40.}$$